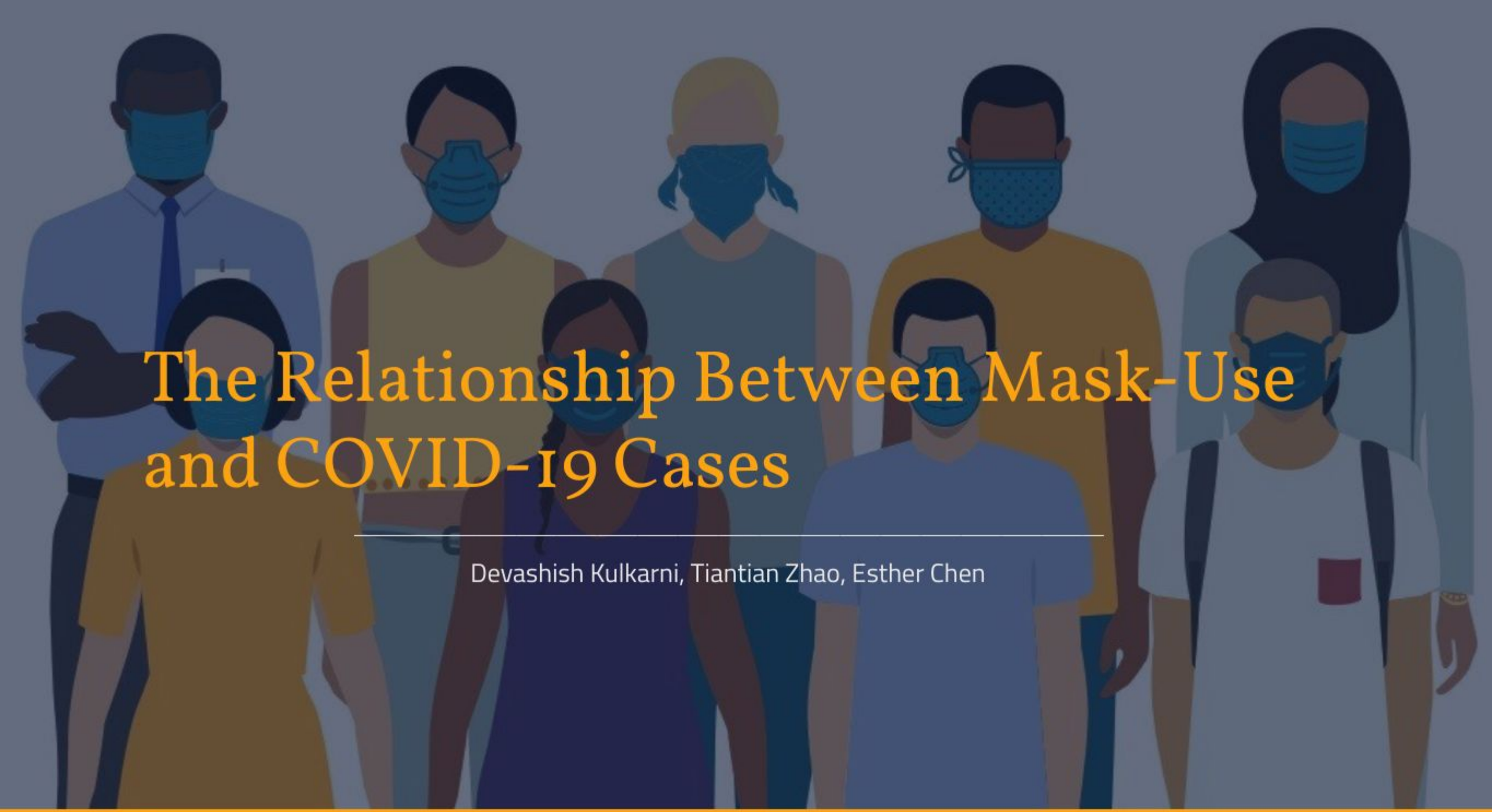


The Relationship Between Mask-Use and COVID-19 Cases

Devashish Kulkarni, Tiantian Zhao, Esther Chen

Mask use behaviors during the Pandemic

Research Question

How does the usage of masks by the general public **affect** the *number of Covid-19 cases* in their respective communities?

Additional Covariates

Urban/Rural

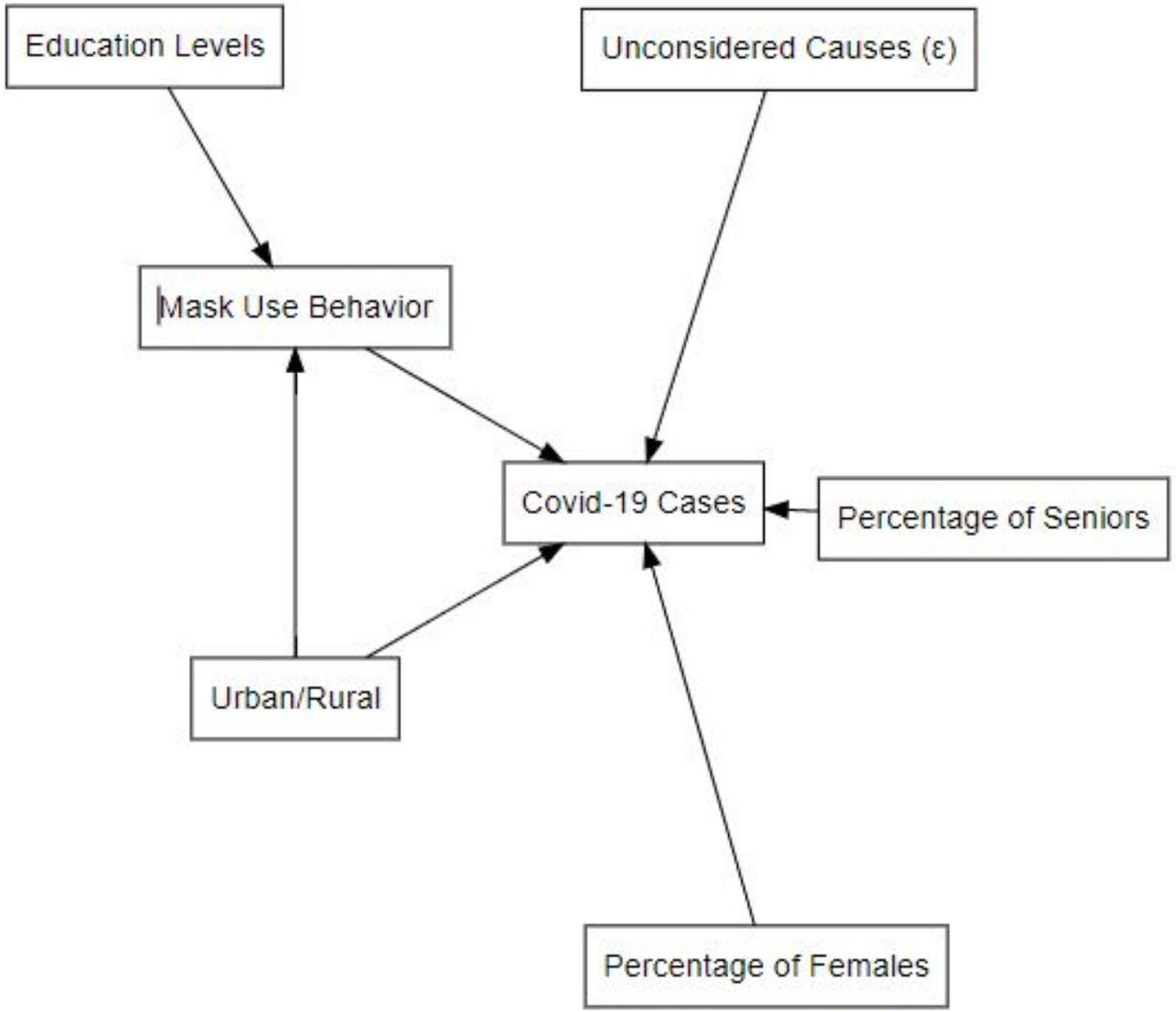
Education level

Percentage of population 60+

Percentage of females

Median age

Causal Graph



Exploratory Data Analysis

Response Variable: COVID-19 cases per hundred thousand residents
(Cases.Per.100K = Cases/Population * 100,000)

Primary Explanatory Variables: Frequency of mask use.
(frequent.mask.use = Frequently + Always)

Covariate Explanatory Variables:

- Overall median age for the county
- The percentage of county population who is 60 or older
- The percentage of adults with a bachelor's degree or higher
- Whether a county is urban or rural

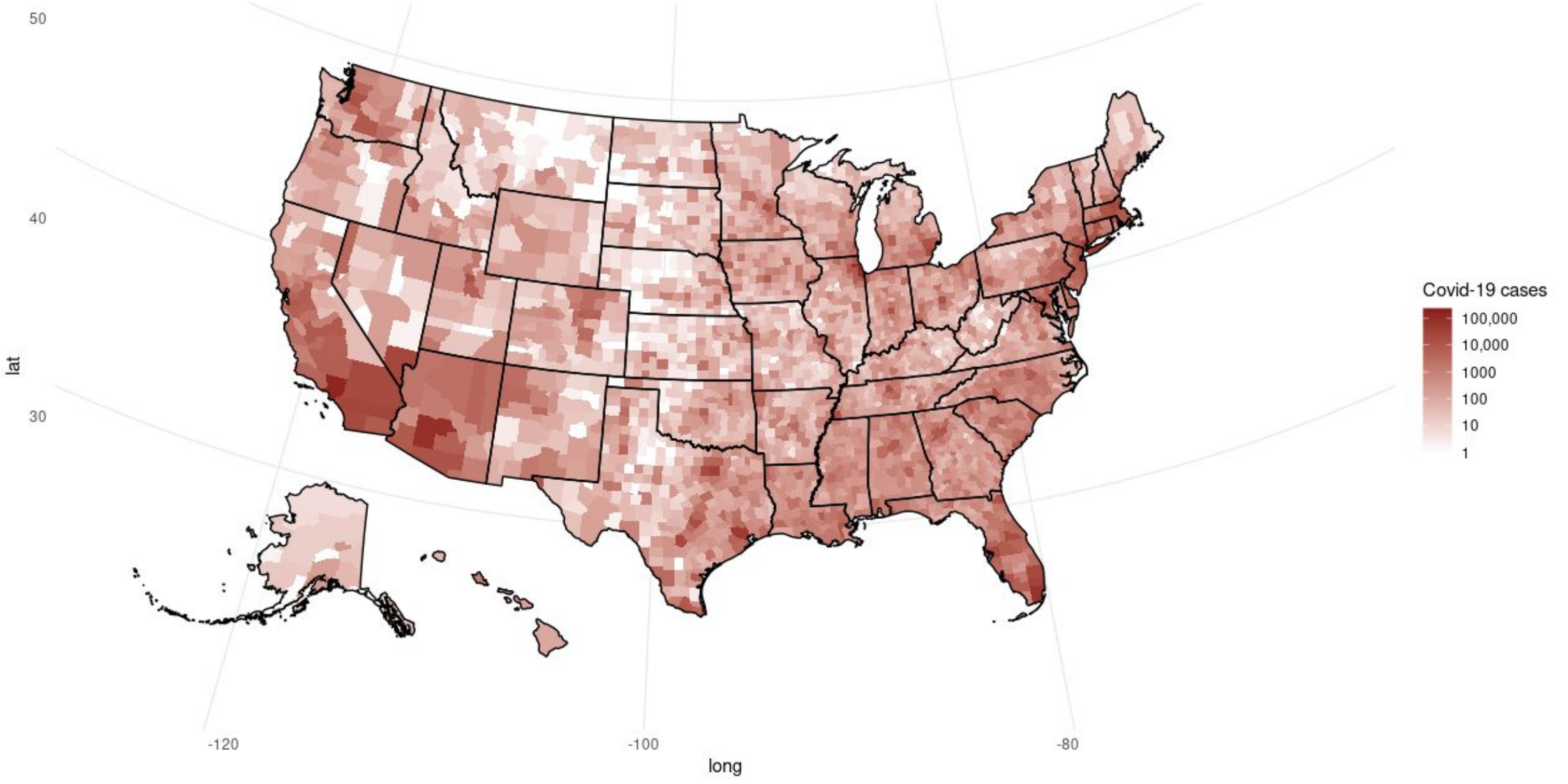

50
40
30
lat
-120
-100
-80
long
Covid-19 cases
100,000
10,000
1000
100
10
1

Figure 1 (a): Histogram of Cases.Per.100K

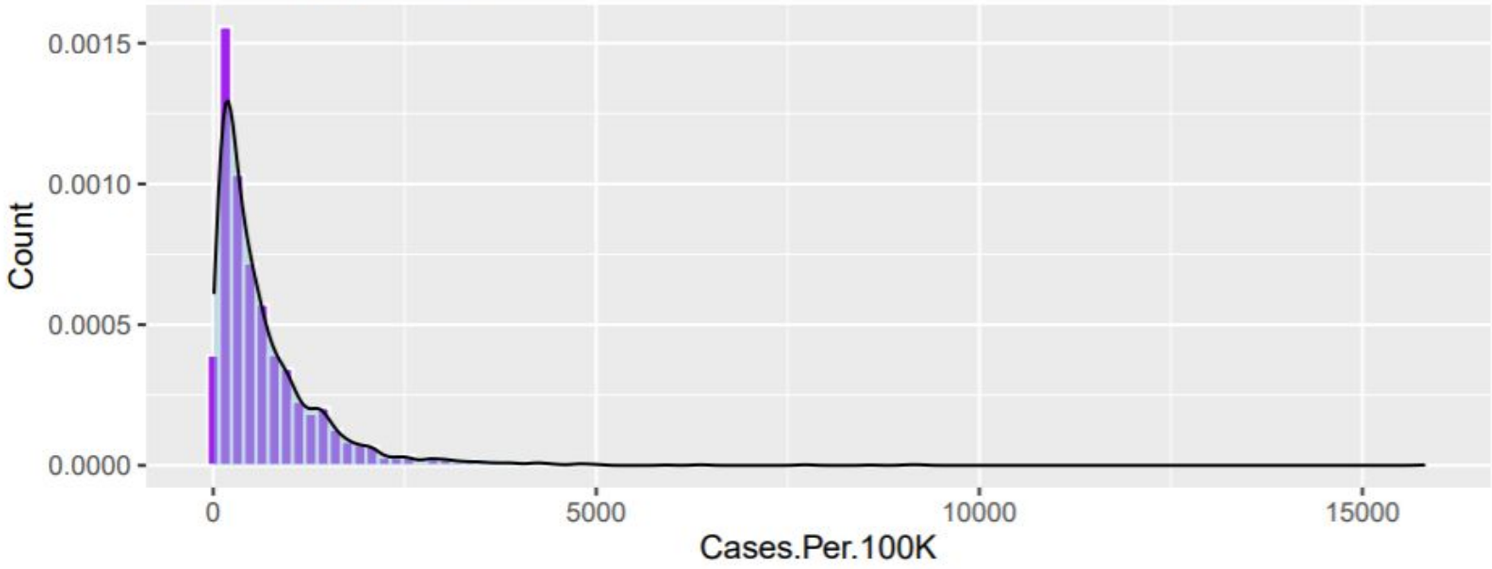


Figure 1 (b): Histogram of log of Cases.Per.100K

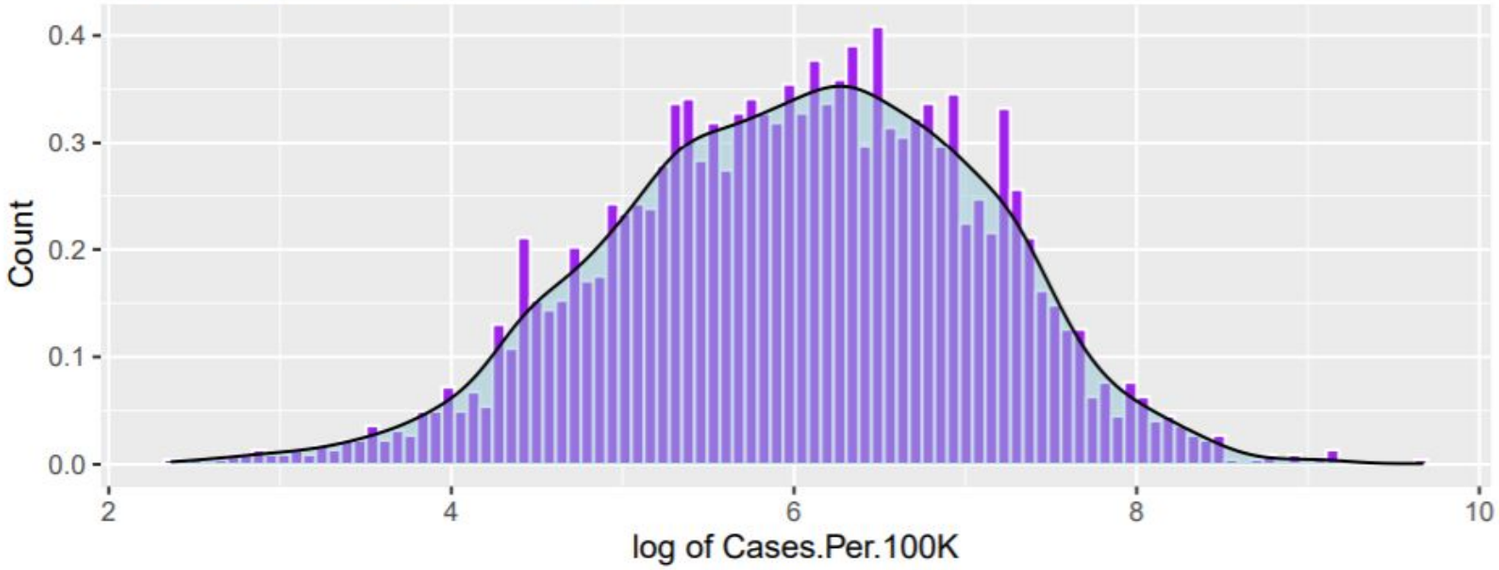
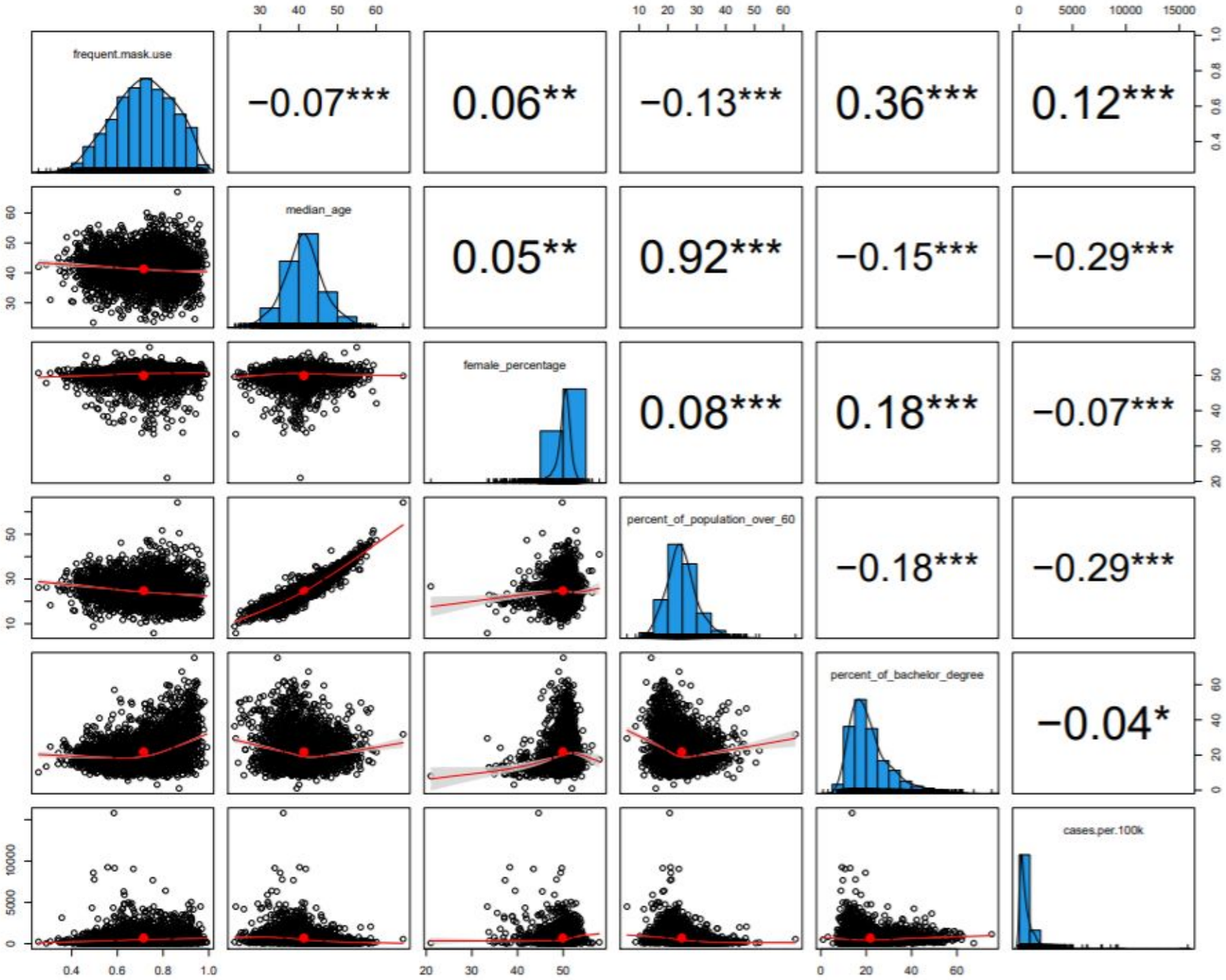


Figure 3: Scatterplot,Histogram and Correlations (All Numeric Varibles)



Model Development

Model 1

$$\text{log of cases per 100k residents} = \beta_0 + \beta_1 * \text{mask.use} + \epsilon$$

Model 2

$$\text{log of cases per 100k residents} = \beta_0 + \beta_1 * \text{mask.use} + \beta_2 * \text{percent.of.population.over.60} + \beta_3 * \sqrt{\text{percent.of.bachelor.degree}} + \beta_4 * urban + \epsilon$$

Model 3

$$\text{log of cases per 100k residents} = \beta_0 + \beta_1 * \text{mask.use} + \beta_2 * \text{median.age} + \beta_3 * \text{female.percentage}^2 + \beta_4 * \text{percent.of.population.over.60} + \beta_3 * \sqrt{\text{percent.of.bachelor.degree}} + \beta_4 * urban + \epsilon$$

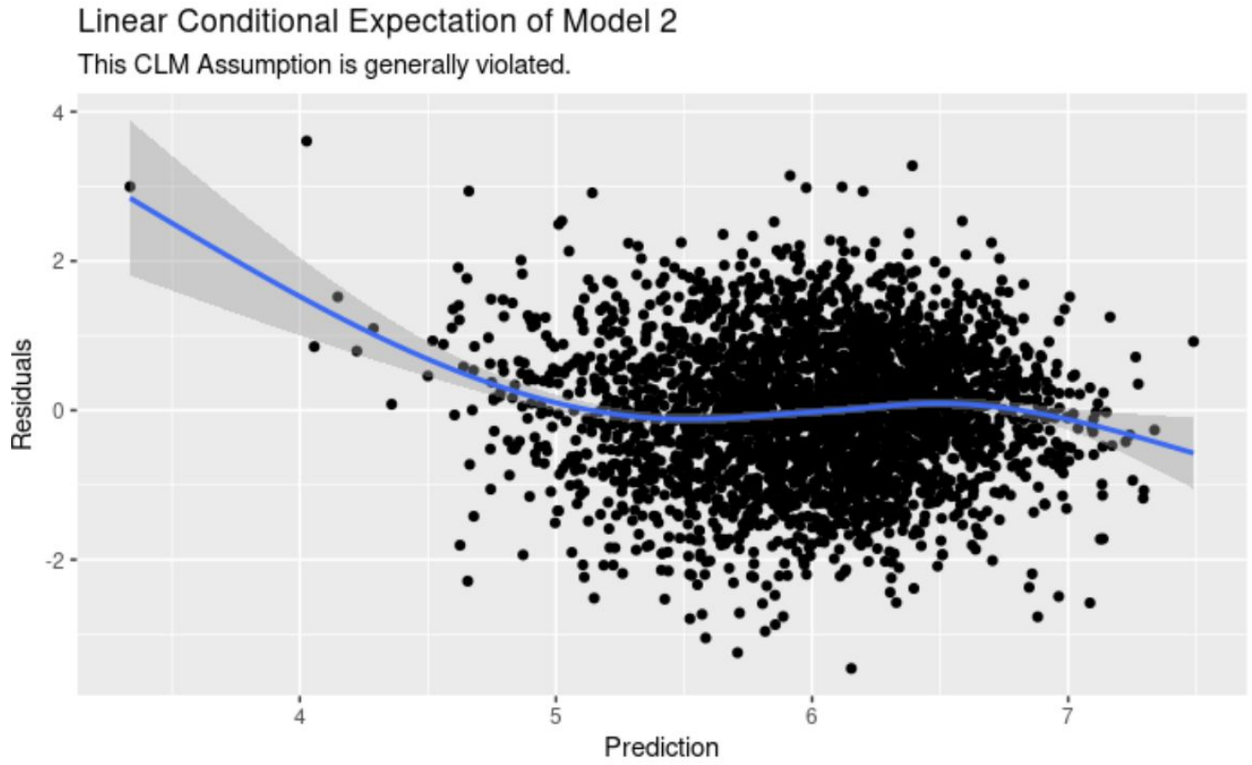
Model Regression Table

```
##
## Model comparison
## ==========================================================================================
##                                                     Dependent variable:
##                                         ----------------------------------------------------
##                                              Reported COVID-19 Cases per 100K Residents
##                                                (1)                (2)                (3)
## ------------------------------------------------------------------------------------------
## Percentage of Residents Wear Masks      1.815***           1.428***           1.526***
##                                          (0.145)            (0.146)            (0.146)
## Median Age for County                                                        -0.043***
##                                                                               (0.009)
## Percentage of Female                                                         0.0004***
##                                                                              (0.0001)
## Percentage of Residents 60+                                -0.073***         -0.036***
##                                                             (0.003)            (0.009)
## Percentage of Residents High Degree                        -0.136***         -0.161***
##                                                             (0.020)            (0.021)
## County (Urban)                                              0.221***          0.231***
##                                                             (0.042)            (0.043)
## Intercept                               4.718***           7.348***           7.215***
##                                          (0.105)            (0.148)            (0.290)
## ------------------------------------------------------------------------------------------
## Observations                              3,021              3,021              3,021
## R2                                        0.049              0.215              0.229
## Adjusted R2                               0.049              0.214              0.228
## Residual Std. Error                 1.038 (df = 3019) 0.943 (df = 3016) 0.935 (df = 3014)
## ==========================================================================================
## Note:                                                         *p<0.1; **p<0.05; ***p<0.01
```

CLM Assumptions

1. Identically Distributed Population (IID) - not fully met

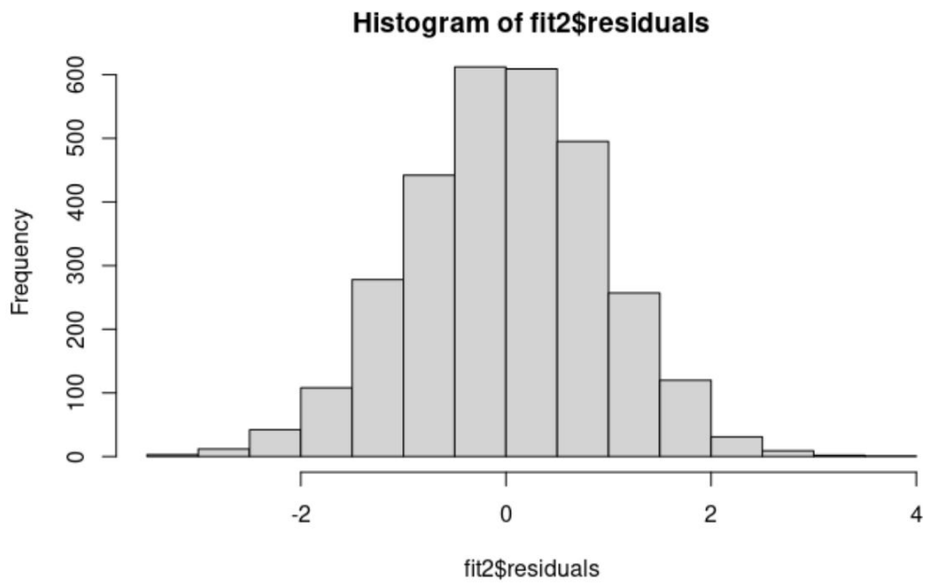
2. Linear Conditional Expectation - violated



3. No Perfect Collinearity - assumption met

4. Homoskedastic Errors - not fully met

5. Normally Distributed Errors - met



Limitations

- Potential reverse causality relationship between mask-use and COVID cases
- Timing of Data Collection (violation of IID)
- Limited test done

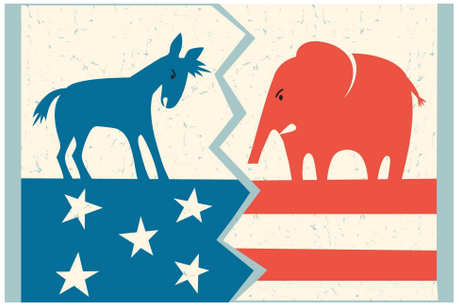
Omitted Variables

1. Transmission Rate

Indicator - If the two indicators suggest different transmission levels, the higher level is selected	Low Transmission Blue	Moderate Transmission Yellow	Substantial Transmission Orange	High Transmission Red
Total new cases per 100,000 persons in the past 7 days	0-9.99	10-49.99	50-99.99	≥100
Percentage of NAATs[1] that are positive during the past 7 days	0-4.99%	5-7.99%	8-9.99%	≥10.0%

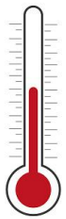
2. Political Views

- Whether a county voted republican or democrat



3. Temperature

- Monthly average temperature of each county



CONCLUSION

THANK YOU

Devashish Kulkarni, Tiantian Zhao, Esther Chen

W203 Summer 2021

Reference

1. "Study shows why the flu likes winter" - The New York Times, https://www.nytimes.com/2007/12/05/health/05iht-05flu.8591550.html
2. CDC Data Tracker, https://covid.cdc.gov/covid-data-tracker/#cases_community

Data Description

Mask use behavior

- Source: New York Times
- Data gathering: Survey (July 2-14, 2020)
- Type: Likert Scale (Never to Always)
- Treatment: Consider Frequent & Always

Data Description

Covid-19 Cases

- Source: New York Times
- Data Gathering: Governments and Health Departments
- Type: Metric
- Treatment: Data from July 14, 2020, normalized by population

Data Description

Education Levels

- Source: Department of Agriculture
- Data Gathering: Compiled from US Census Bureau
- Type: Percentage of adults for various education levels
- Treatment: % completing Bachelor's degree

Data Description

Urban/Rural

- Source: Department of Agriculture
- Data Gathering: Compiled from US Census Bureau
- Type: 12 Categories
- Treatment: 1 & 2 as Urban

Data Description

Percentage of People Above 60

- Source: Kaiser Health News
- Data Gathering: Compiled from US Census Bureau
- Type: Metric
- Treatment: Used as is